Machine Learning with Applications 19 (2025) 100627

Contents lists available at ScienceDirect

Machine Learning with Applications

journal homepage: www.elsevier.com/locate/mlwa

Corrigendum

Corrigendum to "Machine learning for sports betting: should model selection be based on accuracy or calibration?" [Machine Learning with Applications Volume 16, June 2024, 100539]

Conor Walsh *, Alok Joshi

University of Bath

The authors regret that errors in the implementation of our sports betting pipeline were identified during modularisation and unit testing of the code, in anticipation of making it open-source. These errors, occurring in the feature engineering steps of the pipeline, affected all later results. Despite the changes in results, the paper's conclusion remains the same. The following corrections are noted:

(to avoid repetition, we omit text occurring in both error and correction, as signalled by ...)

Highlights

Error: Sports bettors can increase their wealth by a third over a single season.

Correction: Sports bettors can increase their wealth by almost a quarter over a single season

Error: Kelly betting only works with a well-calibrated model.

Correction: Kelly betting can fail even with a well-calibrated model

Abstract

Page 1.

Error: We show that using calibration, rather than accuracy, as the basis for model selection leads to greater returns, on average (return on investment of +34.69% versus -35.17%) and in the best case (+36.93% versus +5.56%).

Correction: We show that using calibration, rather than accuracy, as the basis for model selection leads to greater returns, on average (return on investment of -9.77% versus -26.78%) and in the best case (+23.13% versus +10.9%).

Results

Page 9.

Error: Subset B ... consisted of the following features: 3P, BLK, FT%, ORB, AST%, Previous Season Winning Percentage. Subset C ... consisted of the following features: STL%, eFG%, DRB, AST, DRtg, ORtg, 3P, Previous Season Winning Percentage.

Correction: Subset B ... consisted of the following features: ORtg, DRB, AST. Subset C ... consisted of the following features: ORtg, 3P, Previous Season Winning Percentage, STL%, AST, DRB.

Error: The SVM model was identified as the best calibration-driven model, achieving a classwise-ECE of 3.23% on the test set. This was followed by MLP, with a classwise-ECE of 3.59%, and LR and RF models, with respective classwise-ECEs of 3.61% and 4.39% (see Table 3). The most accurate model was the SVM model with a test set accuracy of 66.55%

…These two final models (calibration-driven SVM and accuracy-driven SVM) were used to generate predictions for the 2018/2019 NBA season.

Correction: The MLP model was identified as the best calibration-driven model, achieving a classwise-ECE of 3.52% on the test set. This was followed by the LR and SVM models, each with a class-wise ECE of 3.77%, and the RF model, with a classwise- ECE of 4.23% (see Table 3). The most accurate model was the SVM model with a test set accuracy of 66.46% ... These two final models (calibration-driven MLP and accuracy-driven SVM) were used to generate predictions for the 2018/2019 NBA season.

Fixed Betting

Error: In this simulation, both systems identified several losing value bets early in the season (see Fig. 3 games 0–20). These losses were not significant, as under the fixed betting rule, the stake was restricted to $100 .... While the pattern of betting appears similar between the two systems, the success enjoyed by each differed significantly. By the halfway-point of the season, the calibration-driven system was in the money, up approximately 25% of its initial budget, while its accuracy-driven counterpart was down approximately 25% (see Fig. 3 circa game 550) .... This difference disappeared in the second half of the season, when the systems appeared to consistently place almost identical bets .... However, the difference over the first half of the season proved decisive, and the gap between the bankrolls remained significant by season's end. The calibration-driven betting system ultimately achieved a highly profitable ROI of 32.45%, while the accuracy-driven system achieved a respectable ROI of 5.56%.

Correction: In this simulation, both systems identified several losing value bets early in the season (see Fig. 3 games 0-100). These series of losses were not disastrous, as under the fixed betting rule, the stake was

DOI of original article: https://doi.org/10.1016/j.mlwa.2024.100539.

* Corresponding author.
E-mail address: conorwalsh206@gmail.com (C. Walsh).

https://doi.org/10.1016/j.mlwa.2025.100627

Available online 8 February 2025

restricted to $100 .... While the pattern of betting appears similar between the two systems (both reaching their lowest bankrolls just after the half-way point of the season, see Fig. 3 circa game 700), the success enjoyed by each differed somewhat. By the thousandth game of the season, the calibration-driven system was in the money, up approximately 10% of its initial budget, while its accuracy-driven counterpart was just shy of breaking even (see Fig. 3 circa game 1000) .... This difference disappeared at the tail end of the season, when the systems appeared to consistently place almost identical bets .... However, the difference over the first 1000 games proved decisive, and the gap between the bankrolls remained significant by season's end. The calibration-driven betting system ultimately achieved a highly profitable ROI of 23.13%, while the accuracy-driven system achieved a respectable ROI of 10.9%

Kelly Betting

Error: Once again, both systems identified several losing value bets early in the season (see Fig. 4 games 0–20) .... This increase in volatility exacerbated the effect on the bankrolls of the qualitative difference in bets identified by each system in the first half of the season. As a result, by the time the accuracy-driven system's bankroll had approximately halved in size, the calibration-driven system's bankroll had increased by the same amount. (see Fig. 4 circa game 500). Volatility continued to be ever-present, and the calibration-driven system increased its bankroll by almost $15,000 (to reach a peak of approximately $20,000) over the span of approximately 100 games (see Fig. 4 circa game 800). While the models identified almost the exact same value bets over the second half of the season (as shown in Fig. 3), the curves look not at all similar towards the end of the season .... The accuracy-driven system consistently diminished as the end of the season approached, ultimately achieving a negative ROI of -75.9%. The calibration-driven system continued to experience high volatility, but mostly remained in the money, and concluded the season with an impressive ROI of 36.93%.

Correction: Once again, both systems identified several losing value bets early in the season (see Fig. 4 games 0-100) .... This increase in volatility exacerbated the effect on the bankrolls of the qualitative difference in bets identified by each system over the first 1000 games of the season. As a result, despite having reached a peak of almost $13,000 (see Fig. 4 circa game 150), by the thousandth game of the season, the accuracy-driven system had lost almost 70% of its initial wealth. The calibration-driven system had not fared much better by this point, and had lost approximately 50% of its initial wealth. However, it did not experience the same volatility throughout the season, and at its peak its bankroll had grown by less than 10% of its initial wealth. As mentioned, the models identified almost the exact same value bets over the tail end of the season (as shown in Fig. 3). While the curves show similarity, they do not quite resemble a mirror image of one another .... The systems continued to experience volatility over the final stretch of the season, and both ultimately lost significant wealth. The calibration-driven system concluded the season with an ROI of -42.67%, and the accuracy-driven system finished with an ROI of -64.45%.

Evaluation of Central Hypothesis

Page 10.

Error: Both models frequently identified value bets, with calibration-driven systems betting on 87.55% of games, and accuracy-driven systems betting on 89.89% of games. Both models also had similar success rates, with the accuracy-driven systems winning 38.46% of bets placed, and the calibration-driven systems winning a slightly higher 38.8% of bets. Unsurprisingly, the accuracy-driven model was the more accurate of the two (accuracy of 64.62% versus 64.27%), while the calibration-driven model was more well-calibrated (classwise-ECE of 4.46% versus 5.03%) .... the maximum ROI achieved by a calibration-driven betting system was 36.93% (achieved using the eighth-Kelly betting rule to determine the stake). The maximum ROI achieved by an accuracy-driven system was 5.56% (under the fixed betting rule). Calibration-driven betting systems were also more profitable on average, with a highly lucrative average ROI of 34.69%, compared to an ROI of -35.17% as achieved by accuracy-driven betting systems on average (see Table 7).

Correction: Both models frequently identified value bets, with calibration-driven systems betting on 82.79% of games, and accuracy-driven systems betting on 88.41% of games. Both models also had similar success rates, with the accuracy-driven systems winning 38.75% of bets placed, and the calibration-driven systems winning a slightly higher 40.23% of bets. Unsurprisingly, the accuracy-driven model was the more accurate of the two (accuracy of 65.14% versus 64.88%), while the calibration-driven model was more well-calibrated (classwise-ECE of 4.59% versus 5.17%) .... the maximum ROI achieved by a calibration-driven betting system was a highly lucrative 23.13% (achieved using the fixed betting rule to determine the stake). The maximum ROI achieved by an accuracy-driven system was 10.9% (under the fixed betting rule). Calibration-driven betting systems were also more profitable on average, with an ROI of -9.77%, compared to an ROI of -26.78% as achieved by accuracy-driven betting systems on average (see Table 7).

Discussion

Page 11.

Error: Basing model selection on calibration led to profitable betting systems in all cases, with an average ROI of 34.69%, and an ROI of 36.93% in the most profitable system. In contrast, basing model selection on accuracy led to an ROI of -35.17% on average and 5.56% in the best case, with the worst-performing system (which used the eighth-Kelly betting rule) losing over 75% of its wealth ... we showed that bettors can increase their wealth by more than a third over a single season.

Correction: Basing model selection on calibration led to an average ROI of -9.77%, and a highly lucrative ROI of 23.13% in the most profitable system. In contrast, basing model selection on accuracy led to an ROI of -26.78% on average and 10.9% in the best case, with the worst-performing system (which used the eighth- Kelly betting rule) losing almost two-thirds of its wealth ... we showed that bettors can increase their wealth by almost a quarter over a single season.

Error: The accuracy-driven SVM identified more value bets than the calibration-driven SVM (betting on 89.89% of games compared to 87.55%) and had a lower success rate (38.46% versus 38.8%).

Correction: The accuracy-driven SVM identified more value bets than the calibration-driven MLP (betting on 88.41% of games compared to 82.79%) and had a lower success rate (38.75% versus 40.23%).

Error: Fig. 5 shows a scatter plot of the model's predicted probability versus the probability implied by the book-maker's odds, for all value bets identified by the calibration-driven SVM, and the accuracy-driven SVM, respectively ... Additionally, this set of predictions exhibits greater variance than its calibration-driven counterpart (0.036 versus 0.034). We can treat the implied probability as a proxy for the true probability (Wheatcroft, 2020). In this case, the accuracy-driven model's predictions appear further from the ground truth, compared to the calibration-driven model.

Correction: Fig. 5 shows a scatter plot of the model's predicted probability versus the probability implied by the book-maker's odds, for all value bets identified by the calibration-driven MLP, and the accuracy-driven SVM, respectively ... This disproportionate concentration of value-bets at the extremes of the probability spectrum may suggest overconfidence on the model's behalf.

Corrected Figures

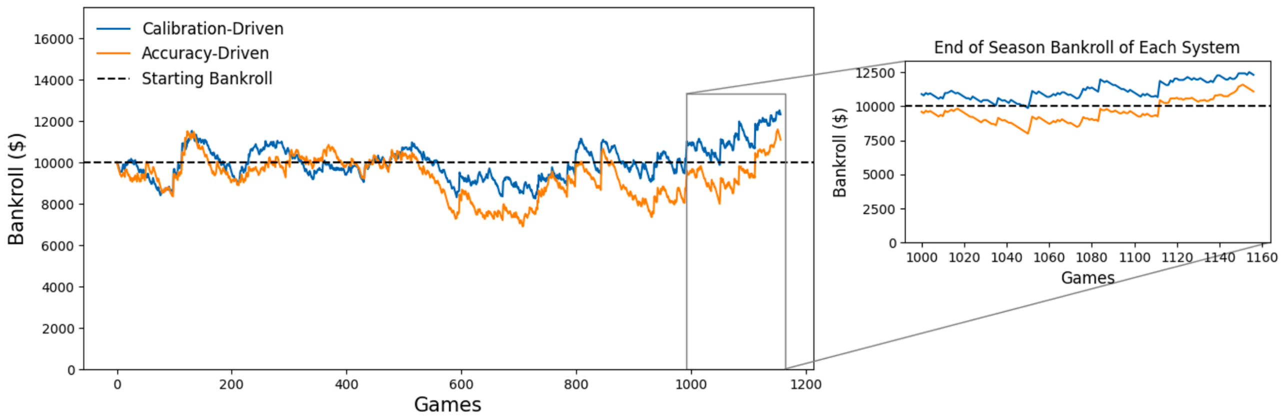
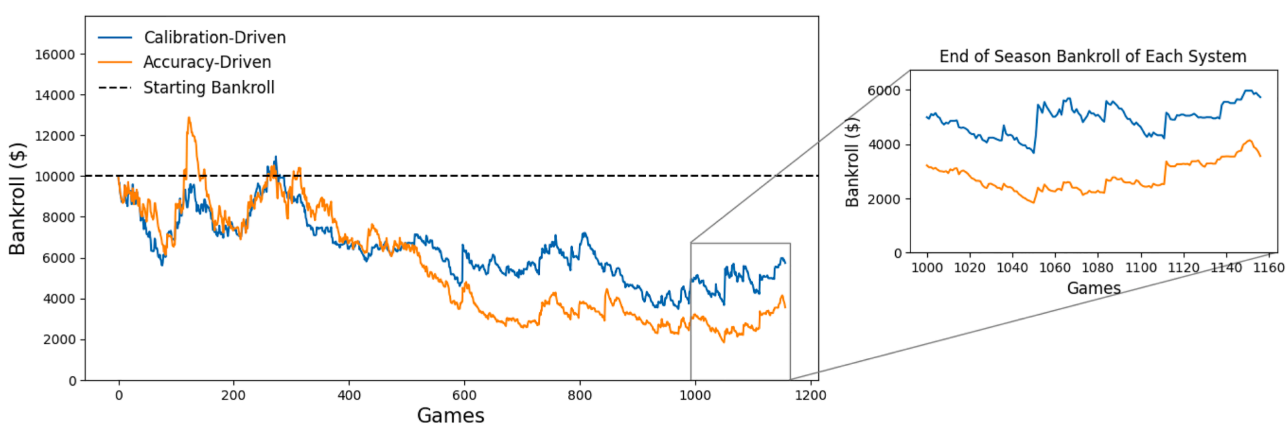
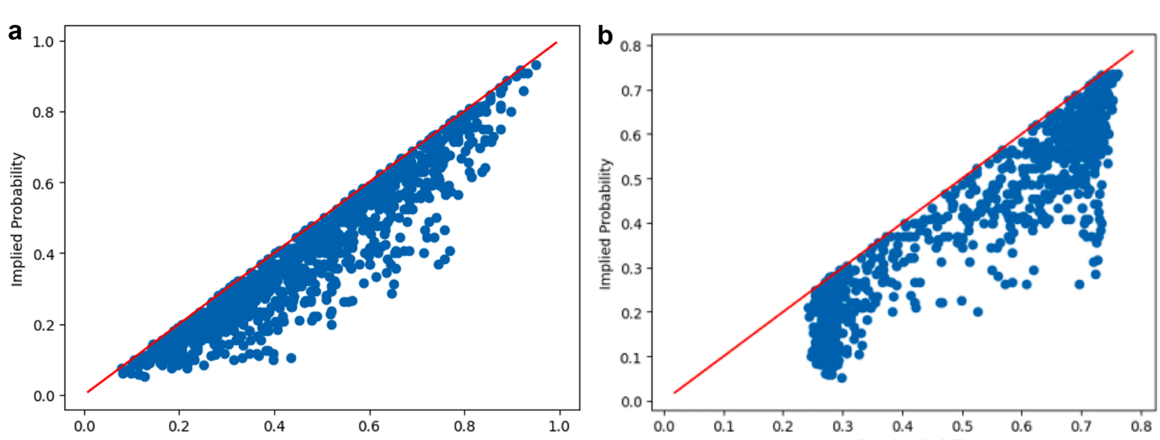


Figure 3: Bankrolls over the course of the 2018/2019 NBA season under the fixed betting rule. The blue curve represents the bankroll of the betting system using the calibration-driven predictive model, while the orange curve represents its accuracy-driven counterpart. The broken black line represents the initial bankroll. Curves which finish above this line earn a profit and are considered successful betting systems.

Figure 4: Bankrolls over the course of the 2018/2019 NBA season under the Eighth-Kelly betting rule. The blue curve represents the bankroll of the betting system using the calibration-driven predictive model, while the orange curve represents its accuracy-driven counterpart. The broken black line represents the initial bankroll. Curves which finish above this line earn a profit and are considered successful betting systems.





(a) Scatter plot of predicted probability versus implied probability for each game identified as a value bet by the calibration-driven MLP.

(b) Scatter plot of predicted probability versus implied probability for each game identified as a value bet by the accuracy-driven SVM.

Figure 5: Comparison of scatter plots showing the predicted probability versus implied probability for each value bet identified by the predictive models (calibration-driven MLP, and accuracy-driven SVM, respectively). All points lie below the line y= x as a value bet is defined as one for which the model's predicted probability is greater than the probability implied by the bookmaker's odds (x > y).

Corrected Tables

Table 3. Classwise-ECE achieved on the test set by each model along the calibration branch

Model	Classwise-ECE
Logistic Regression	3.77%
Random Forest	4.23%
Support Vector Machine	3.77%
Multi-Layer Perceptron	3.52%

Table 4. Accuracy achieved on the test set by each model along the accuracy branch

Model	Accuracy
Logistic Regression	65.34%
Random Forest	65.69%
Support Vector Machine	66.46%
Multi-Layer Perceptron	65.34%

Table 5. Results of fixed betting simulations

Model	Final Bankroll	ROI
Calibration-driven MLP	$12,312.77	23.13%
Accuracy-driven SVM	$11,089.56	10.9%

Table 6. Results of Kelly betting simulations

Model	Final Bankroll	ROI
Calibration-driven MLP	$5,733.04	-42.67%
Accuracy-driven SVM	$3,555.41	-64.45%

Table 7. Comparison of calibration-driven and accuracy-driven betting systems

Model	% Games Bet On	% Bets Won	Accuracy	CLasswise-ECE	Maximum ROI	Average ROI
Calibration-driven MLP	82.79%	40.23%	64.88%	4.59%	23.13%	-9.77%
Accuracy-driven SVM	88.41%	38.75%	65.14%	5.17%	10.9%	-26.78%

Supplementary Material

The updated supplementary material can be found at the following location: https://github.com/conorwalsh99/ml-for-sports-betting.

The authors would like to apologise for any inconvenience caused.